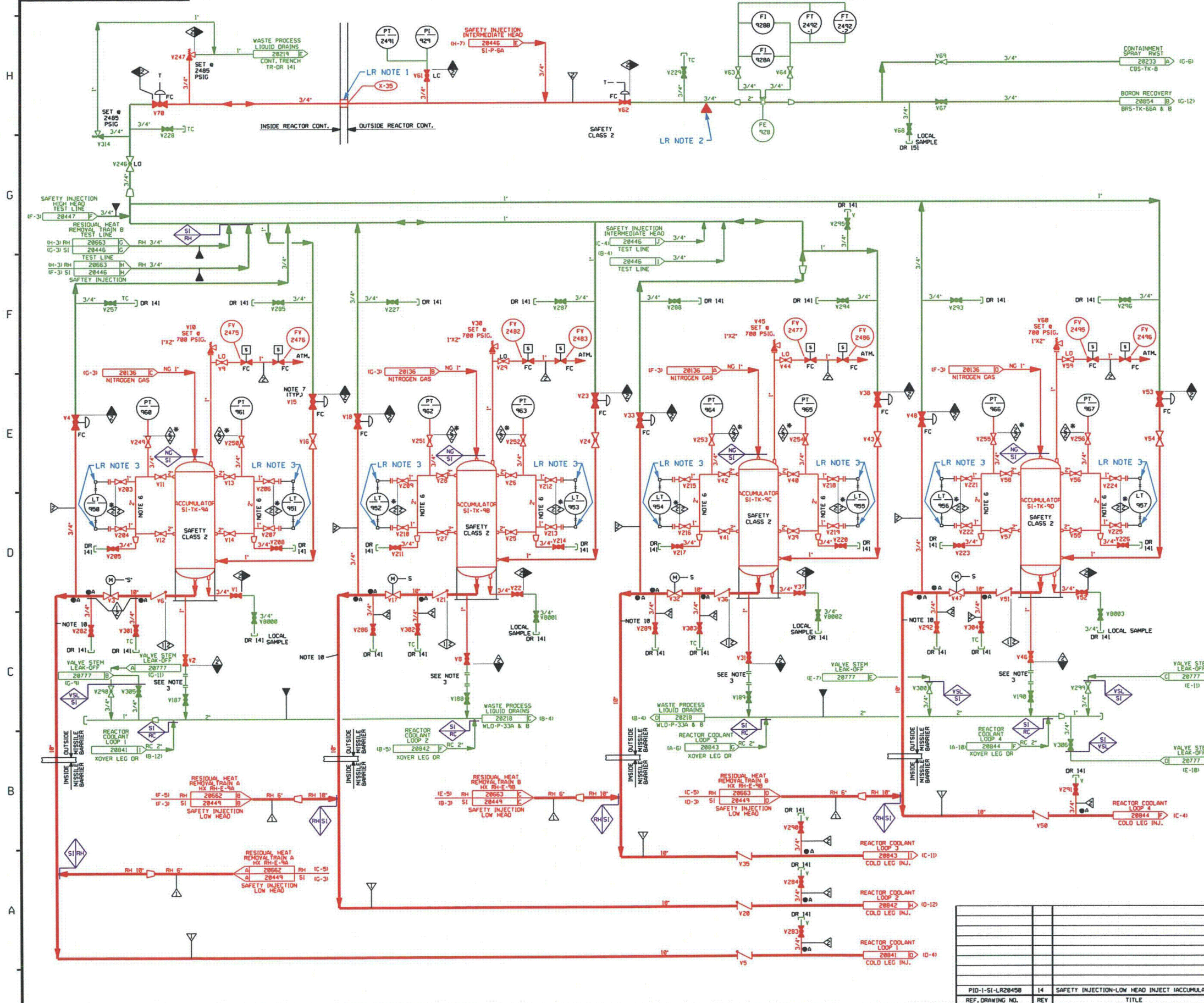


PID-1-SI-LR20450



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

- LR NOTES:
1. COMPONENT DESIGNATED AS X-35 IS A CONTAINMENT PENETRATION THAT IS EVALUATED UNDER THE CIVIL STRUCTURAL SYSTEM LRSP-CNT.
 2. PIPE SUPPORT EQUIVALENT ANCHOR
ISO NO.: 800248-10248
HANGER NOS.: 248-SG-1, 248-RG-4, 248-SG-2 & 248-RG-3A
CALC NO.: MCD 10248-CALC
 3. CAPILLARY TUBING IS PART OF THE INSTRUMENT AND CONSIDERED AN ACTIVE COMPONENT PER NEI 95-10.

COMPONENTS SUBJECT TO AMR
RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

- ▲ PIPE ANCHOR
◆ INDICATES SYSTEM BOUNDARY INTERFACE
SI - SAFETY INJECTION SYSTEM
NG - NITROGEN GAS SYSTEM
RC - REACTOR COOLANT SYSTEM
RH - RESIDUAL HEAT REMOVAL SYSTEM
VSL - VALVE STEM LEAK OFF SYSTEM

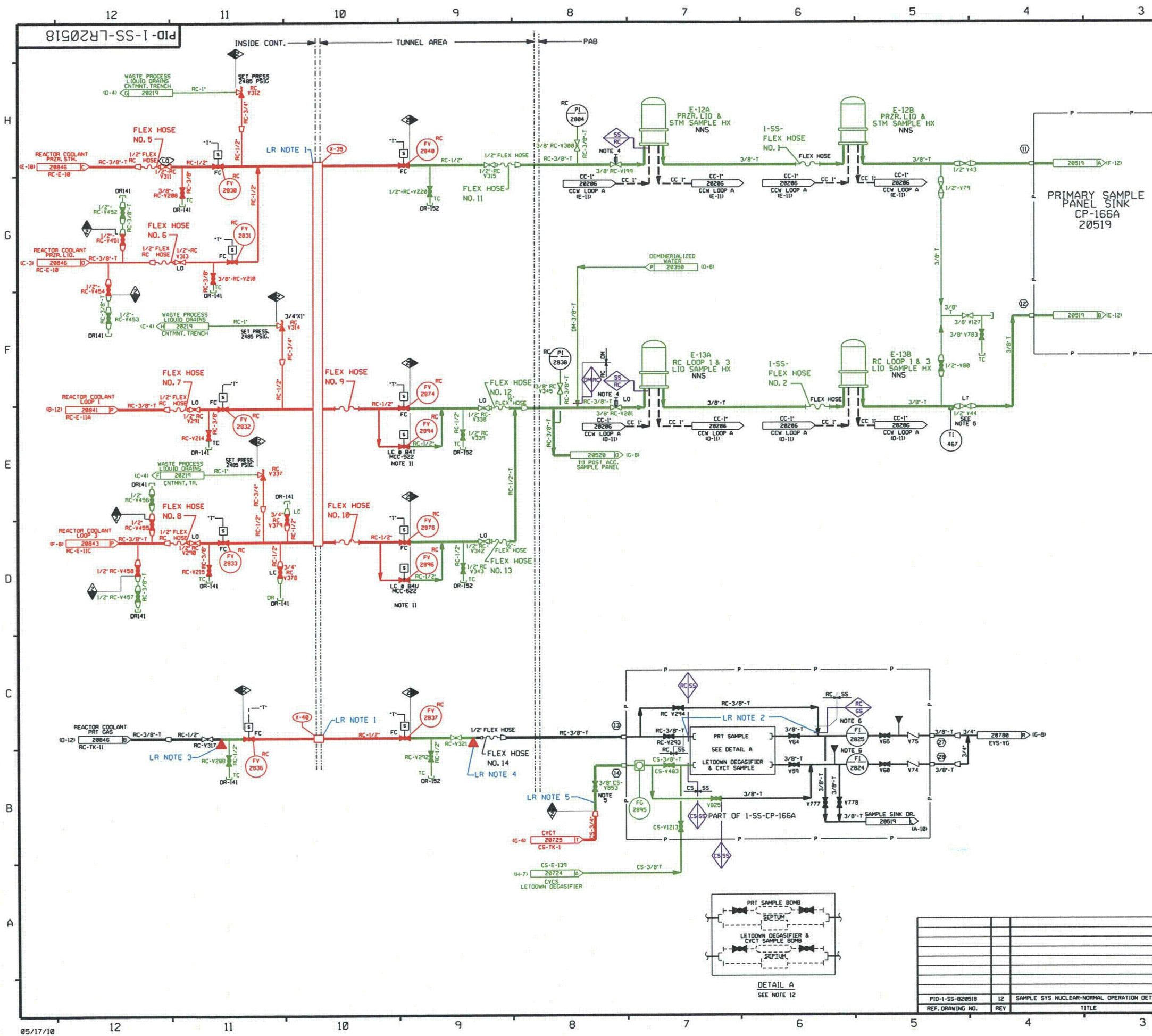
- NOTES:
1. ALL LINES, EQUIPMENT, COMPONENTS, AND INSTRUMENTS HAVE SYSTEM PREFIX SI, UNLESS NOTED OTHERWISE.
 2. WORK THIS DRAWING WITH DRAWINGS 20445 THRU 20449.
 3. ACCUMULATOR DRAIN TO BE PROVIDED WITH BLIND FLANGE, SPOOL PIECE TO BE INSTALLED ONLY WHEN ACCUMULATOR IS TO BE DRAINED.
 4. DELETED
 5. DELETED
 6. 66" MINIMUM STANDPIPE TO BE PROVIDED BY A/E WITH MARK INDICATING NORMAL WATER LEVEL AS DEFINED IN THE PLANT SETPOINT DOCUMENT. LEVEL TRANSMITTER TAPS ARE LOCATED 8 INCHES ABOVE AND BELOW THE SCRIBE MARK.
 7. DELETED
 8. 8" INDICATES 3/8" FLOW RESTRICTOR INSTALLED. SEE APPLICABLE DETAIL ON DWG RC-20848.
 9. DELETED
 10. PIPING SCHEDULE 140 MUST BE ADHERED TO DUE TO SAFETY ANALYSIS FLOW REQUIREMENTS.
 11. DELETED
 12. DELETED
 13. △ INDICATES REVISION LEVEL
 14. VENT, DRAIN & TEST CONN. CODE BREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-MHT-005111 UNLESS NOTED OTHERWISE.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

REV	DATE	DSGN	CHKO	CE	DESCRIPTION
0					ORIGINAL ISSUE
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

SAFETY INJECTION SYSTEM
LOW HEAD INJECTION (ACCUMULATORS)
DETAIL
LICENSE RENEWAL P&ID
PID-1-SI-LR20450

PID-1-SI-LR20450
REF. DRAWING NO. 14
REV. 1
TITLE SAFETY INJECTION-LOW HEAD INJECT (ACCUMULATORS)



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:

- COMPONENTS DESIGNATED AS (X-35 & 40) ARE CONTAINMENT PENETRATIONS THAT ARE EVALUATED UNDER CIVIL/STRUCTURAL SYSTEM LRSP-CNT.
- THIS PORTION OF THE SAMPLE SYSTEM IS NOT IN SCOPE BECAUSE THE INTERNAL ENVIRONMENT IS AIR/GAS.
- PIPE SUPPORT ANCHOR
ISO NO. 800129-492.10A
HANGER NO. 129-A-9
CALC. NO. MCD 492.10A-CALC
- PIPE SUPPORT EQUIVALENT ANCHOR
ISO NO. 800129-10129
HANGER NOS. M/S 129-SG-4A & M/S 129-4A/129-RG-4B
CALC. NO. MCD 10129-CALC
- PER NEI 95-10 APPENDIX F SECTION 4.4F THE SMALLER TUBING DOES NOT IMPOSE LOADS ON THE LARGER PIPING AND DOES NOT SUPPORT THE PIPING.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(1) AND/OR (a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.

BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.

GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.4(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR

◆ INDICATES SYSTEM BOUNDARY INTERFACE

SS - SAMPLE SYSTEM
RC - REACTOR COOLANT SYSTEM
DM - DEMINERALIZED WATER SYSTEM
CS - CHEMICAL & VOLUME CONTROL SYSTEM

NOTES:

- WORK THIS DRAWING WITH DRAWINGS 20516, 20519 THRU 20522.
- ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX SS, UNLESS NOTED OTHERWISE.
- ALL SAMPLE INSTRUMENTATION INDICATION SHALL BE LOCATED ON A PANEL AT THE SAMPLE SINK.
- REACH ROD OPERATED FROM THE SAMPLE ROOM SIDE OF THE SAMPLE ROOM-SAMPLE WALL.
- THIS VALVE SHALL BE ADJUSTED AND SET TO GIVE A 0.8 GPM SAMPLE FLOW FROM RPV LOOP 1 OR 3.
- ALL FLOW INDICATORS ON SS-CP-166A & B HAVE INTEGRAL NEEDLE VALVES WHICH ARE NOT SHOWN ON THE P&ID.
- TYPICAL CONNECTION IDENTIFICATION FOR SS-CP-166A & B
- DELETED
- △ INDICATES REVISION LEVEL
- VENTS, DRAINS AND TEST CONNECTION CODEBREAKS ARE AT THE DOWNSTREAM END OF THE OUTER ISOLATION VALVE PER 1-NHY-005111 UNLESS OTHERWISE NOTED.
- VALVE DISABLED IN THE LOCKED CLOSED POSITION BY LOCKING BREAKERS B4T & B4U OPEN.
- SAMPLE BOMBS AND SEPTUMS ARE INSTALLED DURING SAMPLING FUNCTIONS. EITHER THE SAMPLE BOMB OR THE SAMPLE SEPTUM MAY BE USED. SEE DETAIL A (A-G).

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

REV	DATE	DSGN	CHKD	CE	DESCRIPTION
0					ORIGINAL ISSUE

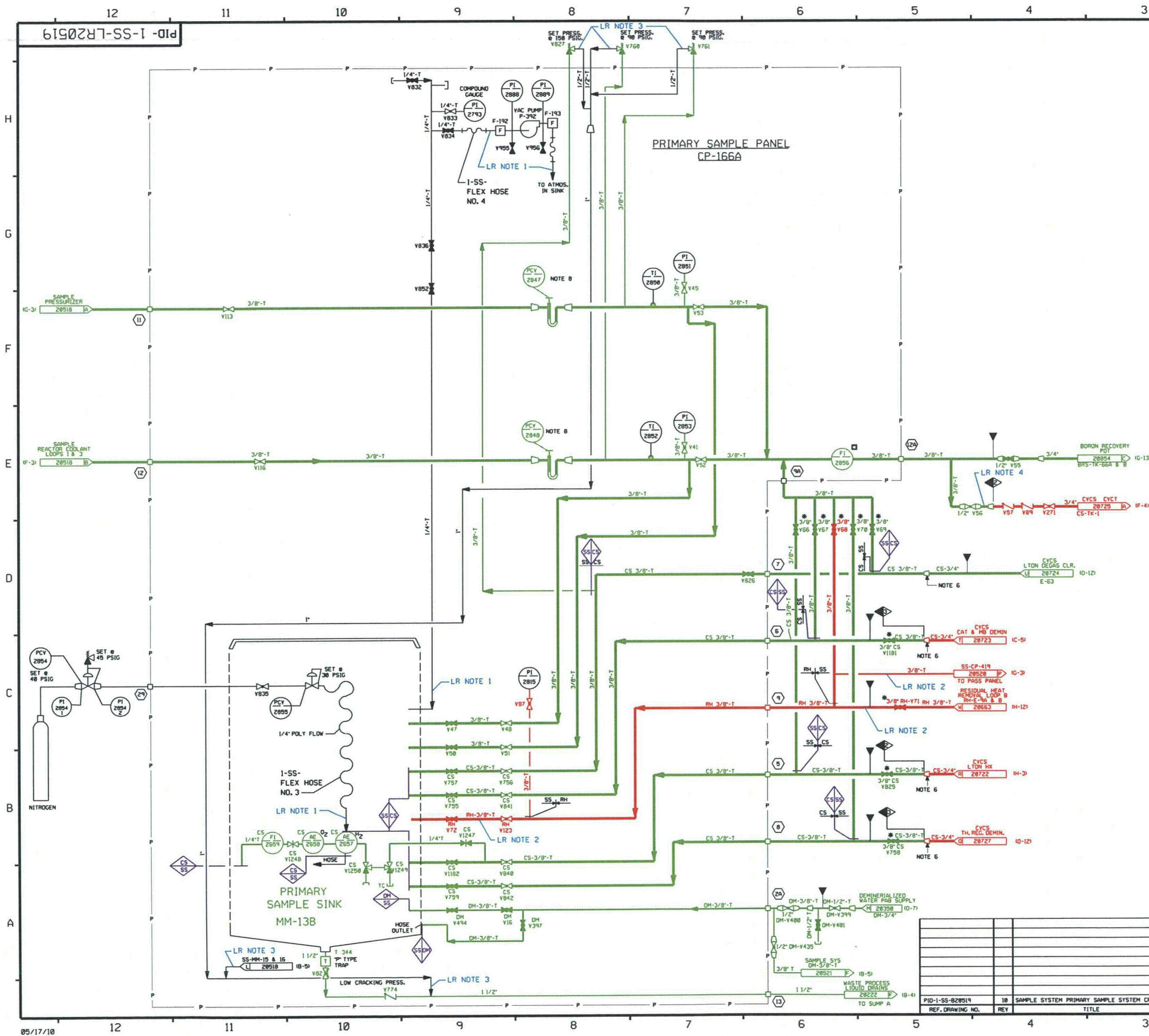
NextEra Energy
SEABROOK

**SAMPLE SYSTEM
(NUCLEAR-NORMAL OPERATION)
DETAIL
LICENSE RENEWAL P&ID**

PID-1-SS-LR20518

REV	DATE	DESCRIPTION
12		SAMPLE SYS NUCLEAR-NORMAL OPERATION DETAIL

PID-1-SS-820518
REF. DRAWING NO.



FOR P&ID REFERENCE DRAWINGS, SYMBOLS AND ABBREVIATIONS
REFER TO DRAWINGS P&ID - LRLEGEND 1 AND P&ID - LRLEGEND 2

LR NOTES:

1. THIS PORTION OF THE SAMPLE SYSTEM IS NOT IN SCOPE BECAUSE THE INTERNAL ENVIRONMENT IS AIR/GAS.
2. PIPING IS IN SCOPE FOR CRITERION 1(a)(3) FIRE PROTECTION.
3. PIPING HAS INTERNAL ENVIRONMENT OF AIR/GAS THEREFORE IS NOT IN SCOPE FOR LICENSE RENEWAL.
4. THE TRANSITION TO TUBING DOES NOT NEED AN EQUIVALENT ANCHOR TO ISOLATE THE CLASS BREAK AT VALVE SS-V-57. REFERENCE CALCULATIONS ARE MCD 565.80-CALC, MCD 10248-CALC AND MCD 503.20-CALC.

COMPONENTS SUBJECT TO AMR

RED INDICATES COMPONENTS IN SCOPE PER 10CFR 54.41(a)(1) AND/OR 1(a)(3) AND SUBJECT TO AMR PER 10CFR 54.21.
BLACK INDICATES COMPONENTS NOT SUBJECT TO AMR.
GREEN INDICATES COMPONENTS IN SCOPE PER 10CFR 54.41(a)(2) AND SUBJECT TO AMR PER 10CFR 54.21.

▲ PIPE ANCHOR



INDICATES SYSTEM BOUNDARY INTERFACE
SS - SAMPLE SYSTEM
DM - DEMINERALIZED WATER SYSTEM
CS - CHEMICAL & VOLUME CONTROL SYSTEM

NOTES:

1. WORK THIS DRAWING WITH DRAWING 28516, 28518, 28520 THRU 28522.
2. ALL LINES, EQUIPMENT, COMPONENTS AND INSTRUMENTS HAVE SYSTEM PREFIX "SS" UNLESS OTHERWISE NOTED.
3. ALL TUBING 3/8" T AND MAT'L 2588 UNLESS OTHERWISE NOTED.
4. DELETED
5. DELETED
6. THESE REDUCERS ARE TO BE AS CLOSE AS POSSIBLE TO ROOT VALVE.
7. * THIS VALVE TO BE REACH ROD OPERATED FROM THE SAMPLE RM SIDE OF THE SAMPLE RM ROOM - SAMPLE WALL.
8. THIS EQUIPMENT HAS 1/4" TUBING CONNECTORS.
9. △ INDICATES REVISION LEVEL.
10. DELETED.
11. ○ CONTINUATION CONNECTION SEE FP-74882.

PORTIONS OF THIS DRAWING ARE
NUCLEAR SAFETY RELATED

REV	DATE	DSGN	CHKD	CE	DESCRIPTION
1					ORIGINAL ISSUE
SAMPLE SYSTEM PRIMARY SAMPLE PANEL CP-166A LICENSE RENEWAL P&ID PID-1-SS-LR20519					

